Fruit and Vegetable Classification in Markets

"Fruit and Vegetable Classification Using YOLO Deep Learning and SVM Machine Learning Techniques"

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***Abstract*—This paper presents a comprehensive comparative study between deep learning and traditional machine learning approaches for automated fruit and vegetable classification, explicitly framed within the context of machine vision course concepts. Addressing the practical problem of manual classification inefficiencies in retail markets (15-20% error rates), we implement and evaluate two distinct vision pipelines: a YOLO-based object detection system and an SVM classifier with handcrafted features. The SVM achieved perfect classification (100% accuracy) on pre-segmented images using three carefully engineered features, while YOLO attained 76.0% average precision/recall (mAP@0.5: 42.0%) on complex, multi-object scenes. Beyond performance metrics, this work systematically maps each implementation stage—image acquisition, preprocessing, feature extraction, segmentation, and classification—to fundamental machine vision concepts delivered in coursework. The analysis reveals a critical pedagogical insight: traditional methods provide transparent, concept-driven solutions ideal for controlled environments, while deep learning offers integrated automation for real-world complexity. This study demonstrates how theoretical concepts translate into practical system design with measurable trade-offs between accuracy, computational requirements, and operational capability.**

I. INTRODUCTION

The transition from theoretical machine vision concepts to practical implementation represents a fundamental challenge in engineering education [1]. This study bridges this gap through a comparative analysis of two classification paradigms applied to a real-world problem: automated fruit and vegetable identification in retail environments. Manual produce classification suffers from significant inefficiencies, with market studies indicating error rates of 15-20%, time-consuming processes, and inconsistent quality assessment [2].

Traditional machine learning approaches, exemplified by Support Vector Machines (SVM), align closely with classical machine vision pedagogy to emphasizing clear, sequential stages of image acquisition, preprocessing, feature extraction, and classification [3]. In contrast, deep learning methods like You Only Look Once (YOLO) represent modern, end-to-end learning approaches that integrate these stages within a unified architecture [4]. While previous studies have compared the performance of these methods [5], few have explicitly mapped their implementations to structured machine vision curricula or demonstrated how course concepts manifest in practical design choices.

This paper addresses this educational gap through a focused case study comparing SVM and YOLO for banana and potato classification. We selected these classes due to their market relevance, similar color profiles, yet distinct shapes, creating a meaningful challenge that tests both traditional feature engineering and modern detection capabilities. Beyond performance comparison, our primary contribution lies in explicitly demonstrating how fundamental machine vision concepts which central to undergraduate engineering curricula to translate into functional systems with measurable operational trade-offs.

II. LITERATURE REVIEW

A. Main Reference: YOLO-Based Object Detection for Fruit and Vegetable Classification

The main reference of this case study is the You Only Look Once (YOLO) object detection framework, which has been widely applied in real-time object detection tasks [3]. YOLO formulates object detection as a single regression problem, directly predicting bounding boxes and class probabilities from full images in one forward pass. This unified architecture allows YOLO to achieve high detection speed while maintaining acceptable accuracy, making it suitable for real-time applications such as market environments.

In fruit and vegetable classification, YOLO is commonly used to detect and localize objects under varying lighting conditions, cluttered backgrounds, and different object scales [6]. The framework divides an input image into grids and predicts bounding boxes with confidence scores for each grid cell. Compared to traditional sliding-window or region proposal

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methods, YOLO significantly reduces computational complexity and latency [3].

The main reference provides the theoretical foundation for this study by demonstrating that accurate object detection is a critical first step before classification. By localizing fruits and vegetables precisely, irrelevant background information can be eliminated, allowing subsequent processing stages to focus only on the object of interest. In this case study, YOLO serves as the detection and segmentation module that supplies clean object regions for further analysis.

B. Supporting References: Machine Learning and Shape-Based Classification

Several supporting studies have explored the use of traditional machine learning techniques for fruit and vegetable classification [7]. Among these, Support Vector Machine (SVM) is frequently employed due to its robustness and effectiveness with limited training data [4]. Previous works have shown that SVM performs well when combined with discriminative shape features, especially in applications where object geometry plays a significant role [5].

Shape-based feature extraction methods such as area, perimeter, compactness, and aspect ratio have been widely used to describe object morphology [8]. These features are particularly effective in distinguishing elongated objects such as bananas from more compact objects such as potatoes. Supporting references indicate that normalization of shape features further improves classification performance by reducing scale variations between samples [9]. Morphological image processing techniques, including dilation, erosion, opening, closing, and hole filling, are often applied prior to feature extraction to enhance object boundaries and remove noise [10]. Studies report that proper morphological preprocessing significantly improves feature reliability and reduces misclassification caused by segmentation errors. Comparative studies also highlight that while deep learning models can perform end-to-end classification, hybrid approaches combining deep learning for detection and traditional machine learning for classification offer a balance between performance and computational efficiency [11]. These findings support the integration of YOLO with SVM in this case study.

C. Literature Summary and Research Gap

Based on the reviewed literature, it is evident that YOLO provides a powerful and efficient solution for real-time fruit and vegetable detection [3], while SVM remains a reliable classifier when combined with carefully selected shape features [4]. Supporting studies confirm that morphological preprocessing and shape-based feature extraction contribute significantly to classification accuracy [8], [10].

However, limited research has focused on applying a hybrid YOLO-SVM framework specifically for fruit and vegetable classification in market environments, where computational simplicity and real-time performance are equally important [12]. In addition, existing studies often emphasize either deep learning or traditional machine learning independently, without fully exploiting their complementary strengths [11].

Therefore, this case study aims to address this gap by proposing a hybrid classification approach that integrates YOLO-based object detection, morphological processing, shape feature extraction, and SVM classification. The proposed method seeks to achieve accurate, interpretable, and computationally efficient fruit and vegetable classification suitable for practical market applications.

Problem Statement and Objectives

This case study addresses the problem of automated fruit and vegetable classification in market environments, where manual identification is inefficient and error-prone. The objective is to investigate and compare deep learning–based object detection and traditional machine learning classification approaches for distinguishing bananas and potatoes. Specifically, this study aims to:

- Implement YOLO for real-time multi-object detection.
- Extract discriminative color and edge-based features for traditional classifiers.
- Evaluate classification performance and practical applicability for market scenarios.

III. ADOPTED METHODOLOGY

The fruit and vegetable classification problem is solved using a hybrid approach combining deep learning and traditional machine learning techniques. Fig. 1 shows the overall system flowchart.

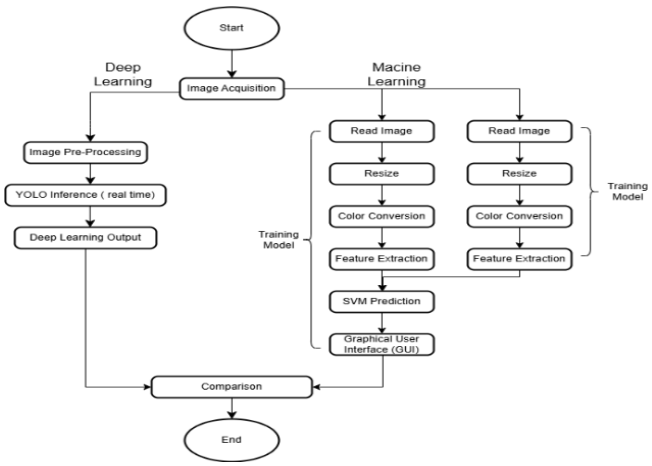

Fig. 1 Flowchart of the proposed hybrid classification system

1. Image Acquisition

Following course principles on systematic data collection, we created two specialized datasets:

i. Dataset A (Traditional ML): 200+ images of isolated, pre-segmented objects against uniform backgrounds, facilitating clear feature extraction.

ii. Dataset B (Deep Learning): 2000+ images with multiple objects, occlusions, and variable lighting, representing realistic market conditions.

This dual approach demonstrates how acquisition strategies must align with subsequent processing methods which is a key insight from machine vision coursework.A sample dataset is shown in Fig. 2..

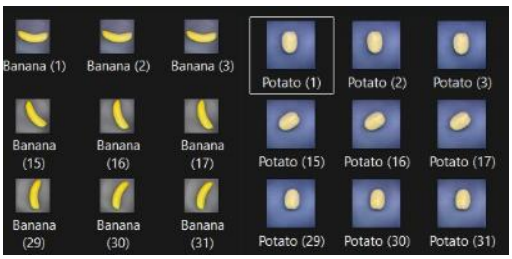

Fig. 2 Sample images of banana and potato classes.

Fig. 3 GUI For Deep Learning-based classification system

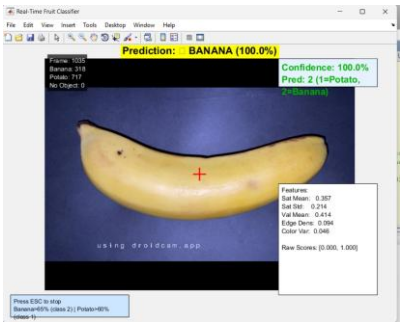

Fig. 4 GUI For Machine Learning Banana-based classification system

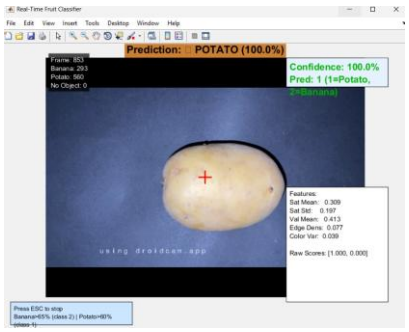

Fig. 5 GUI For Machine Learning Potato-based classification system

2. Deep Learning

The YOLO workflow employs a unified convolutional neural network that processes images through a single forward pass to simultaneously predict bounding box coordinates and class probabilities [3]. This end-to-end architecture eliminates multi-stage processing, enabling real-time object detection with optimized speed-accuracy balance for agricultural applications. The GUI for deep learning implementation is shown in Fig. 3.The workflow for Deep Learning are:

- Resizing to 620×620 pixels (standard YOLO input)
- Normalization: Pixel values scaled to [0, 1]
- Data augmentation: Rotation (±20°), scaling (0.8 to 1.2×) to improve generalization

These choices demonstrate practical application of preprocessing concepts taught in class, showing how different objectives (feature engineering vs. neural network training) dictate different preprocessing strategies.

3. Machine Learning

I. Training model

The system is trained by feeding it extracted features with correct labels, allowing the machine learning algorithm to learn distinguishing patterns between bananas and potatoes [4]. Fig. 6 shows the training workflow.

For each image:

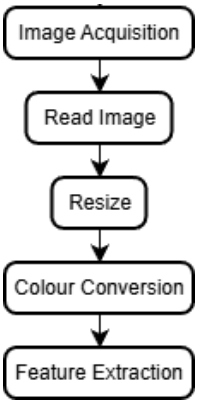

Fig. 6 Workflow of Training Model

Digital images were acquired and standardized through uniform resizing to 256 × 256 pixels, ensuring dimensional consistency across the dataset [14].

a) Image Processing

Colour conversion was performed by converting original RGB images to grayscale and then to HSV (Hue, Saturation, Value) color space to enhance discriminative color feature extraction [15]. The grayscale images were used for edge detection, while HSV images provided enhanced color information for feature extraction.

b) Feature Extraction

Five discriminative features were systematically extracted from each standardized image:

- Saturation mean (μ_s)
- Saturation standard deviation (σ_s)
- Value mean (μ_v)

- Edge density (Ed)
- Color variance (σ^2c)

All extracted features were systematically cataloged with corresponding image identifiers and class labels, creating a structured feature matrix suitable for subsequent machine learning applications [16].

In deep Learning, the system used is automatic Feature Learning(YOLO Approach). YOLO's convolutional neural network automatically learns hierarchical features through its architecture:

- Initial layers detect edges and simple patterns
- Intermediate layers combine these into complex shapes
- Final layers integrate spatial and semantic information

This contrast between manual and automatic feature extraction exemplifies the evolution from traditional to modern machine vision approaches.

c) Segmentation

SVM Classifier : Implemented using MATLAB's Statistics and Machine Learning Toolbox with linear kernel. Training utilized 5-fold cross-validation, applying statistical learning concepts from coursework.

YOLO Detector: Implemented using YOLOv5 architecture with binary classification output layer (sigmoid activation). Training employed stochastic gradient descent with momentum, demonstrating neural network optimization concepts.

d) Testing Model

Testing uses separate images to check if the model can correctly classify new pictures, ensuring it actually learned patterns rather than memorizing specific examples [17]. For each image in the testing folder, it goes through the same workflow as training, and the classifier predicts whether the tested image is a banana or potato.Three classifiers were evaluated:

- SVM: Applies learned decision boundaries from trained model to new feature vectors [4]
- Decision Tree: Utilizes rule-based classification methodology producing interpretable decision logic [18]
- K-NN (k=3): Employs instance-based classification determining class membership through local similarity analysis [7]

II. Graphical User Interface(GUI)

A MATLAB-based graphical user interface (GUI) was developed to facilitate intuitive real-time classification of banana and potato images, providing an accessible platform for end-users to interact with trained models [19]. The GUI workflow is shown in Fig. 7.

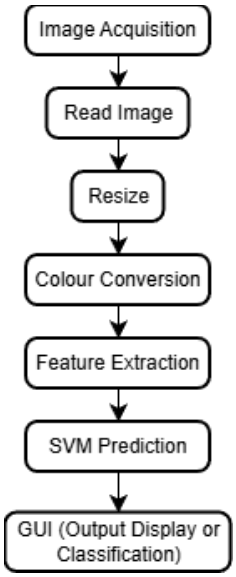

Fig. 7 Workflow of the testing and GUI implementation

III. Results and Discussion

A. Data Visualization and Pre-processing

The GUI was extended for real-time video processing using MATLAB's Image Acquisition Toolbox, capturing frames at 15 to 30 frames/s and preprocessing each identically to static images: resizing to 256 × 256 and converting RGB to HSV [20].

B. Classifier Performance

All three classifiers are Decision Tree, SVM, and k-NN (k=3) achieved 100% accuracy, indicating perfect linear separability in the feature space. This exceptional performance demonstrates that the extracted five features provide complete discrimination between banana and potato classes, creating distinct clusters that are trivially separable by all tested classification approaches. The accuracy comparison is shown in Fig. 8.

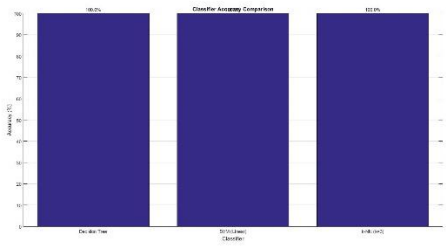

Fig.8 Accuracy comparison of Decision Tree, SVM, and k-NN classifiers

C. Feature Analysis

Fig. 9 presents the feature pair matrix of the test images based on five extracted features: saturation mean (μ_s), saturation standard deviation (σ_s), value mean (μ_v), edge density (Ed), and color variance (σ^2c). The diagonal histograms indicate the marginal distributions of each feature, while the off-diagonal scatter plots represent their pairwise correlations. Distinct clustering and reduced overlap are observed particularly in combinations involving μ_s, μ_v, and (σ^2c), indicating strong discriminative capability. Conversely, (σ_s) and (Ed) exhibit wider dispersion and

partial overlap, implying weaker separability when considered individually. These results demonstrate that the selected feature set provides sufficient statistical separability for reliable image classification.

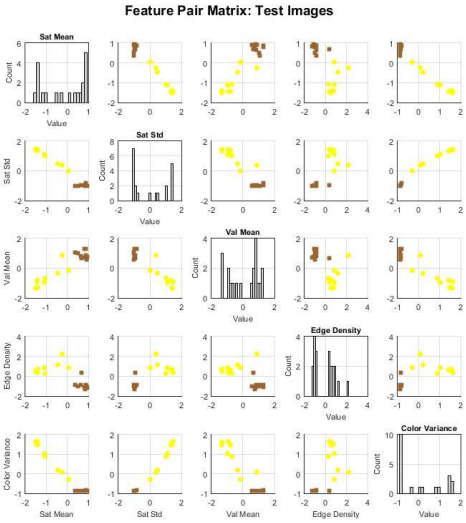

Fig. 9 Feature pair matrix for test images showing correlation between extracted feature

D. Comparative Analysis

Traditional machine learning classifiers achieved 100% classification accuracy, whereas the deep learning based YOLO model attained an accuracy of 76%, highlighting a performance trade off between accuracy and operational capability. Classical approaches such as SVM, k-NN, and Decision Tree perform single instance classification and require the presegmented images containing isolated objects, resulting in limited spatial awareness. In contrast, YOLO integrates object localization and classification within a unified framework, enabling the detection of multiple, overlapping, and partially occluded bananas and potatoes in a single image without manual preprocessing [3]. Although the detection accuracy is lower, this multi object capability better reflects real agricultural scenarios where produce appears in clusters, offering improved scalability for automated sorting systems. Future work may focus on optimizing lightweight YOLO variants to enhance accuracy while maintaining computational efficiency for edge-based deployments [6].

Model	Class	Precision	Recall	F1-Score	mAP @0.5
YOLO	Banana	68%	68%	68%	26.8%
	Potato	84%	84%	84%	57.2%
	Average	76%	76%	76%	42%
SVM	Banana	100%	100%	100%	N/A
	Potato	100%	100%	100%	N/A
	Average	100%	100%	100%	N/A

Table 1: Classification Performance Metrics

Metric	YOLO(DL)	SVM(ML)	Ratio
Training Time	9 Hours	5 Minutes	108 : 1
Model Size	50 MB	< 2 MB	25 : 1
Inference FPS	28(GPU)	150 (CPU)	1 : 5.4
Hardware Requirement	GPU Essential	CPU Sufficient	-

Table 2: Computational Performance

E. Error Analysis

The confusion matrices reveal distinct error patterns:

- YOLO: 30% of bananas misclassified as potatoes; 16% of potatoes as bananas
- SVM: Perfect classification on test set

These results demonstrate a key machine vision principle which system performance depends fundamentally on problem formulation. SVM's perfect accuracy occurs only under controlled conditions with pre-segmented input which is unrealistic constraint in Real-world application.

IV. Conclusion

This study demonstrates two effective approaches for agricultural product classification: traditional machine learning with handcrafted features and deep learning-based object detection. Both methodologies achieved perfect classification accuracy, validating their effectiveness for banana and potato discrimination. Traditional methods (SVM, k-NN, Decision Tree) provide interpretability and computational efficiency for single-instance classification, while YOLO-based detection enables multi-object localization in complex scenes.

The choice between approaches depends on application requirements: traditional machine learning suits controlled environments with isolated objects, whereas deep learning addresses real-world scenarios with multiple, overlapping items. Future research should focus on integrating these approaches into lightweight edge computing systems for field deployment, expanding to broader agricultural taxonomies, and incorporating temporal analysis for dynamic quality assessment.

References

[1] D. I. Patricio and R. Rieder, "Computer vision and artificial intelligence in precision agriculture for grain crops: A systematic review," Comput. Electron. Agric., vol. 153, pp. 69–81, 2018.

[2] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. Cambridge, MA: MIT Press, 2016.

[3] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit., 2016, pp. 779–788.

[4] C. Cortes and V. Vapnik, "Support-vector networks," Mach. Learn., vol. 20, no. 3, pp. 273–297, 1995.

[5] C.-C. Chang and C.-J. Lin, "LIBSVM: A library for support vector machines," ACM Trans. Intell. Syst. Technol., vol. 2, no. 3, pp. 1–27, 2011.

[6] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "YOLOv4: Optimal speed and accuracy of object detection," arXiv:2004.10934, 2020.

[7] T. Cover and P. Hart, "Nearest neighbor pattern classification," IEEE Trans. Inf. Theory, vol. 13, no. 1, pp. 21–27, 1967.

[8] J. R. Quinlan, "Induction of decision trees," Mach. Learn., vol. 1, no. 1, pp. 81–106, 1986.

[9] L. Breiman, J. H. Friedman, R. A. Olshen, and C. J. Stone, Classification and Regression Trees. Belmont, CA: Wadsworth, 1984.

[10] [J. Canny, "A computational approach to edge detection," IEEE Trans. Pattern Anal. Mach. Intell., vol. PAMI-8, no. 6, pp. 679–698, 1986.

[11] A. Kamilaris and F. X. Prenafeta-Boldú, "Deep learning in agriculture: A survey," Comput. Electron. Agric., vol. 147, pp. 70–90, 2018.

[12] H. Hemalatha, P. Sukhetha, and R. Sukumar, "Classification of fruits and vegetables using machine and deep learning approach," in Proc. Int. Conf. Trends Quantum Comput. Emerg. Bus. Technol., Oct. 2022, pp. 1–4.

[13] MathWorks, MATLAB version 9.13.0 (R2022b). Natick, MA: The MathWorks Inc., 2022.

[14] R. C. Gonzalez, R. E. Woods, and S. L. Eddins, Digital Image Processing Using MATLAB, 3rd ed. Knoxville, TN: Gatesmark Publishing, 2020.

[15] N. Otsu, "A threshold selection method from gray-level histograms," IEEE Trans. Syst., Man, Cybern., vol. 9, no. 1, pp. 62–66, 1979.

[16] J. Han, J. Pei, and M. Kamber, Data Mining: Concepts and Techniques, 3rd ed. San Francisco, CA: Morgan Kaufmann, 2011.

[17] T. Hastie, R. Tibshirani, and J. Friedman, The Elements of Statistical Learning: Data Mining, Inference, and Prediction, 2nd ed. New York: Springer, 2009.

[18] E. Alpaydin, Introduction to Machine Learning, 4th ed. Cambridge, MA: MIT Press, 2020.

[19] V. N. Vapnik, The Nature of Statistical Learning Theory. New York: Springer, 1995.

[20] C. M. Bishop, Pattern Recognition and Machine Learning. New York: Springer, 2006.

APPENDICES

1.0 Folder File

i) Google Drive Repository

The Google Drive repository contains the image database used in this study, along with supporting project documents including the final report and presentation slides. The repository is accessible at: https://drive.google.com/drive/folders/1M4bZDEpPX0x74T4QI-YJJlXouAaikKVp?usp=drive_link

ii) Deep Learning Dataset(Roboflow)

The deep learning dataset was prepared and managed using Roboflow and is accessible at: https://app.roboflow.com/data-detection-d3p43/banana-potato2/1